BLOCK DIAGRAM ALGEBRA: CLOSED-LOOP P SPEED CONTROL

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1.0 Learning Outcomes

Upon completing this practical exercise, students should be able to demonstrate the Learning Outcomes presented in Table 1. At the end of the exercise, revisit each Learning Outcome and confirm that it has been achieved.

Table 1: *Learning Outcomes for the Practical Exercise*

ID	Learning Outcome	Achieved
1	Reduce the block diagram of a proportional speed control loop to a single closed-loop transfer function.	☐
2	Predict the closed-loop gain and time constant for several proportional gains, and measure them on the DC motor.	☐
3	Identify where the prediction and the measurement disagree, and explain which modelling assumption causes the difference.	☐

2.0 Hardware Set-Up

This practical exercise builds on the hardware and algorithm set-up described in the practical exercise, DC Motor Speed Measurement using an Encoder, which details the operation of a DC motor with an encoder and the measurement of motor speed in revolutions per minute (RPM). The set-up is not repeated here.

The first-order model of the motor, $G(s) = K/(\tau s + 1)$, identified in the practical exercise, Parameter Estimation of a DC Motor using Linear Least Squares, is used throughout. Use the values of K and τ identified for your own motor. Where example values are needed, this guide uses the A1.4 example results, $K = 28$ RPM/V and $\tau = 0.15$ s.

3.0 Practical Investigation via Simulink

MATLAB and Simulink should be used to complete this practical exercise. A proportional controller is placed in a unity-feedback loop with the motor, the block diagram is reduced to a single transfer function, and the predicted closed-loop gain and time constant are checked on the hardware. The exercise supports the ACE-CORE learning guide, Block Diagram Algebra.

Reduce the Block Diagram

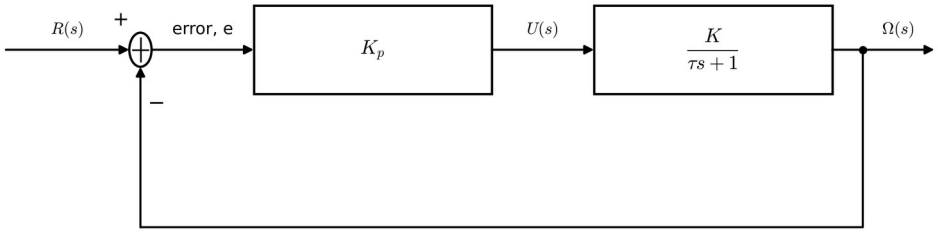


Figure 1: *Block Diagram of Proportional Speed Control with Unity Feedback*

In Figure 1, the controller K_p and the motor $G(s)$ are in series, and the loop has unity feedback. The standard feedback reduction, $T(s) = G_f/(1 + G_f)$, where $G_f = K_pG(s)$ is the forward path, gives:

$$T(s) = \frac{\Omega(s)}{R(s)} = \frac{K_pK}{\tau s + 1 + K_pK} = \frac{K_{cl}}{\tau_{cl}\, s + 1}$$

$$K_{cl} = \frac{K_pK}{1 + K_pK}, \qquad \tau_{cl} = \frac{\tau}{1 + K_pK}$$

The closed loop is still first order. Increasing K_p makes it faster and brings its gain closer to 1, but the gain never reaches 1, so proportional control alone always leaves an error.

1. Reduce the block diagram by hand to obtain $T(s)$, K_{cl} and τ_{cl}.
2. Choose three values of K_p so that the loop gain K_pK is 0.5, 1 and 2. For the example motor these are K_p = 0.018, 0.036 and 0.071 V/RPM.
3. Predict K_{cl}, τ_{cl} and the final speed for a 60 RPM setpoint at each gain, and enter them in Table 2.

Measure the Closed-Loop Response

[Screenshot to insert] *Step (setpoint, RPM) → Sum (+ −) → Gain Kp → Saturation 0–5 V → Gain 255/5 → PWM output; encoder speed (RPM) → low-pass filter → feedback to the Sum. Scopes log r, y, u and t.*

Figure 2: *Simulink Model for Proportional Speed Control*

1. Build the model in Figure 2, with a Saturation block limiting the controller output to 0–5 V before the Gain of 255/5.
2. Apply a setpoint step from 0 RPM to 60 RPM at t = 0.5 s and log the setpoint, speed, controller output and time.
3. Run the test for each value of K_p, and measure the final speed and the time constant (63.2% of the final change).

4. Check whether the controller output reached the 5 V limit in any test.

Example results are shown in Figure 3. They are simulated with the motor model identified in the practical exercise, Parameter Estimation of a DC Motor using Linear Least Squares (K = 28 RPM/V, τ = 0.15 s), so they show the behaviour to expect rather than measured data. The final speed rises towards the setpoint and the response gets faster as K_p increases, as the reduced transfer function predicts.

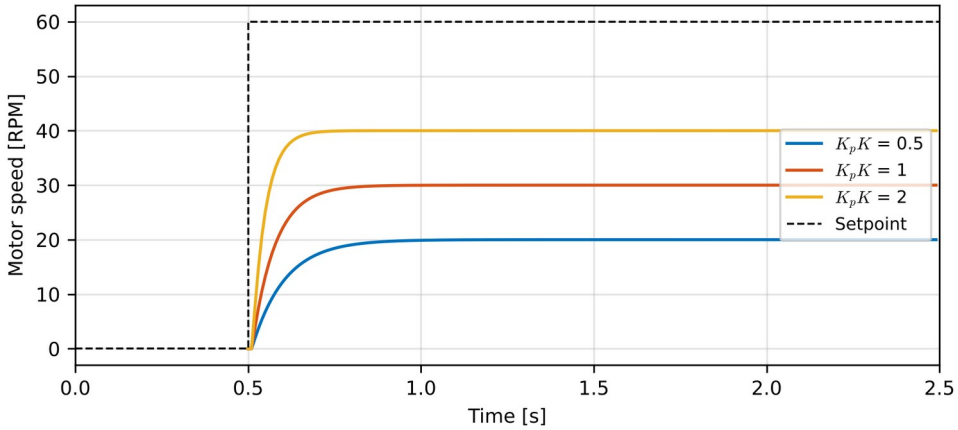


Figure 3: *Simulated Example of the Closed-Loop Response to a 60 RPM Setpoint for Three Proportional Gains*

Table 2: *Predicted and Measured Closed-Loop Responses*

K_p (V/RPM)	K_pK	K_{cl}	τ_{cl} (s)	Predicted final (RPM)	Measured final (RPM)	Measured τ_{cl} (s)
0.018	0.5	0.333	0.100	20.0		
0.036	1	0.500	0.075	30.0		
0.071	2	0.667	0.050	40.0		

The predicted values in Table 2 use the example model. Replace them with predictions from your own K and τ.

1. At which gains does the measurement agree well with the prediction? Where does it break down, and which assumption of the linear model fails there?
2. Why can proportional control alone never bring this motor exactly to the setpoint?
3. Redraw the block diagram with a low-pass filter $1/(\tau_f s + 1)$ in the feedback path and reduce it to a single transfer function. What order is the closed loop now?

4.0 Concluding Remarks

This practical exercise reduced the block diagram of a proportional speed control loop to a single closed-loop transfer function and tested its predictions on

hardware. Using MATLAB, Simulink and ACE-Lab hardware, the closed-loop gain and time constant were predicted and measured for three proportional gains.

Higher gains made the loop faster and reduced the steady-state error, as the reduced transfer function predicts. Disagreement appeared where the linear model stops being valid, such as when the controller output saturates or the motor operates near its dead-zone, which shows both the power and the limits of block diagram analysis.

Recommended Citation

If you have used this practical exercise in your teaching, learning or research, please cite it using the reference below, adapting the format as required by your preferred referencing style.

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